RESEARCH COMPILATION 2026

Intelligence Research Atlas

The corpus map, the five load-bearing connections across domains, and reading paths. Companion to five domain volumes.

Intelligence Research Atlas - Volume 0 of 6 -

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About this atlas

This volume is the entry point to a six-volume research atlas compiled in October 2026: 138+ landmark papers across five domains - the mathematics of intelligence, the deep learning canon, large language model architecture and training, the neuroscience of intelligence, and the frontiers beyond current LLMs. Each domain volume contains the full distilled findings and its paper canon. This overview volume presents the corpus map, the five load-bearing cross-domain connections, and recommended reading paths.

The one-paragraph summary

Intelligence - biological or artificial - appears to be compression. Brains and neural networks both build predictive world models, and both improve by minimizing prediction error: Shannon information, Friston free energy, and cross-entropy loss are formally kin. Scale matters through smooth power laws (Chinchilla), yet qualitatively new abilities appear at thresholds (emergence, contested); reasoning emerged again in 2025 when reinforcement learning was pointed at verifiable rewards (DeepSeek-R1), making test-time compute a second scaling axis. Neuroscience contributes the blueprints backpropagation lacks - recurrence for memory, dopamine as a prediction-error signal, replay for consolidation, sparse distributed codes, and hierarchy as a predictive engine - while mathematics supplies the invariant patterns (power laws, criticality, symmetry, renormalization, information bottlenecks) that recur across brains, networks, and nature.

The five load-bearing connections

If you remember nothing else from 138 papers, remember these five bridges between fields. Each is developed fully in the cross-domain map of the master atlas.

1. Compression is the universal objective

Shannon entropy, LLM cross-entropy loss, and cortical free energy are the same quantity minimized by the telegraph engineer, the model trainer, and (per Friston) the brain. Domains 01, 03, 04.

2. Dopamine is the TD error - and RL is the brain's evaluation system

The reward-prediction-error algorithm formalized in Q-learning, scaled with human preferences (RLHF) and verifiable rewards (RLVR) to produce reasoning. Its pathologies - reward hacking, over-optimization - mirror biological addiction. Domains 02, 03, 04.

3. Memory is an active process, not a log

Hippocampal maps, grid codes, and prioritized replay are the blueprint for agent memory: episodic stores, spatial indexing, consolidation phases. Experience replay in DQN was the first loan; agent systems still under-collect. Domains 04, 05.

4. Search plus learned priors beats either alone

AlphaGo's template - learned evaluation with explicit search at decision time - returned as chain-of-thought, Tree of Thoughts, and reasoning models, and points toward planning inside learned world models. Domains 02, 03, 05.

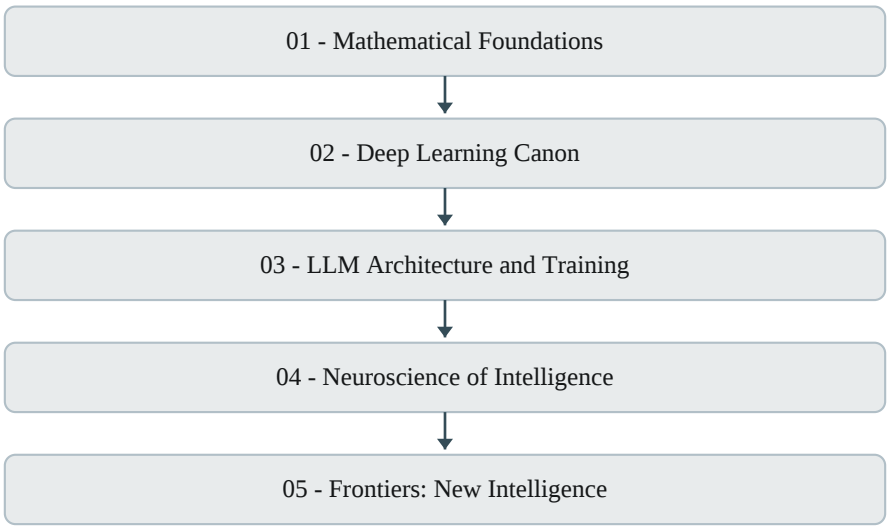
5. Sparse distributed coding, twice over

V1 sparse codes and transformer superposition independently converged on packing more concepts than units via sparse geometry; both needed dictionary methods to decode what was stored. Domains 03, 04.

Corpus map

Five volumes, one per domain; this chain shows the dependency order in which the domains inform each other.

The Intelligence Research Atlas - five domains



Reading paths

A - Build or fine-tune LLMs

03 findings, then 01 F8-F10, targeted 03 papers, 05 F11-F12.

B - Design a new architecture

01 F1-F5/F11, all of 04, all of 05, cross-domain map.

C - Neuroscience foundation

04 in full, then 01 F11/F14 for the mathematical mirrors.

D - Alignment and safety

03 F3/F6/F12, 05 F12-F13, debate / weak-to-strong / Illusion of Thinking.

E - The whole story chronologically

timeline, then 02, 03, 01, 04, 05.

How to use this corpus (agents and humans)

The corpus root contains AGENTS.md (navigation protocol for AI agents), INDEX.md (master index), index/papers-master.json (machine-readable paper index), and one folder per domain with papers.md, findings.md, papers.json, and Mermaid diagrams that render in Obsidian and GitHub. PDFs are build artifacts: regenerate with tools/build_pdfs.py after editing content.